

Booting Procedure of RHEL7/8 and Troubleshooting

Q.1 Reboot your CentOS 7 or RHEL 8 virtual machine and, during startup, press the appropriate key to access the GRUB2 boot menu; take a screenshot of the GRUB menu screen.

Ans- Step 1- Restart VM

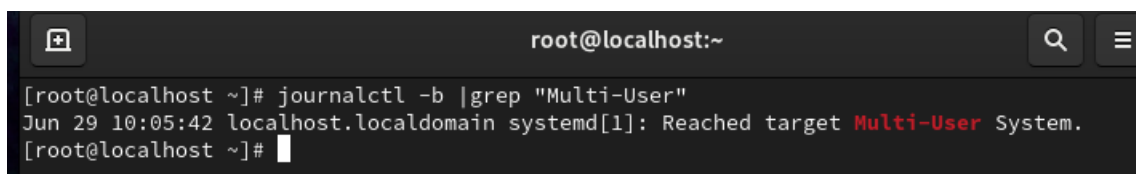
Step 2- Boot time press Esc(UEFI) Or Shift(BIOS) press



Q.2 Use the 'journalctl -b' command to view the system logs from the current boot and identify the timestamp when the multi-user.target was reached.

Hint: Search for 'Reached target Multi-User System' in the output.

Ans- Step -Multi user Target timestamp: journalctl -b | grep "Multi-User"



Q.3 Simulate a boot failure by editing the /etc/fstab file and adding a fake mount entry, then reboot your VM and document the error message that appears during boot.
Constraint: Restore your original /etc/fstab after testing to avoid permanent boot issues.

Ans- Step 1- Backup: `cp /etc/fstab /etc/fstab.bak`

Step 2- Check: `ls -l /etc/fstab*`

Step 3-Edit in fstab: `vi /etc/fstab`

Step 4 Add this line in fstab: `/dev/fake/test ext4 defaults 0 0`

Step 5- Reboot

```
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# ls -l /etc/fstab*
-rw-r--r--. 1 root root 613 Jun 29 10:26 /etc/fstab
[root@localhost ~]# cp /etc/fstab /etc/fstab.bak
[root@localhost ~]# ls -l /etc/fstab /etc/fstab.bak
-rw-r--r--. 1 root root 613 Jun 29 10:26 /etc/fstab
-rw-r--r--. 1 root root 613 Jun 29 10:28 /etc/fstab.bak
[root@localhost ~]# cp /etc/fstab.bak /etc/fstab
```

Q.4 Use the 'systemctl list-units --failed' command after booting to identify any failed services, and write down the name of one failed service and the command you would use to check its status.

Ans- Step 1-Failed services: `systemctl list-units --failed`

Step 2-Check Status: `systemctl status sshd`

```
[root@localhost ~]# systemctl list-units --failed
UNIT LOAD ACTIVE SUB DESCRIPTION
0 loaded units listed.
[root@localhost ~]# systemctl status <service-name>
bash: syntax error near unexpected token `newline'
[root@localhost ~]# systemctl status sshd
• sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena>
   Active: active (running) since Mon 2026-06-29 10:05:27 EDT; 26min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 908 (sshd)
     Tasks: 1 (limit: 10738)
    Memory: 1.7M
       CPU: 27ms
    CGroup: /system.slice/ssh.service
            └─908 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jun 29 10:05:27 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jun 29 10:05:27 localhost.localdomain sshd[908]: Server listening on 0.0.0.0 po>
Jun 29 10:05:27 localhost.localdomain sshd[908]: Server listening on :: port 22.
Jun 29 10:05:27 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```

Q.5 Suppose your system is stuck at the emergency shell prompt during boot. Use ChatGPT to generate a step-by-step recovery plan for this scenario, then try following those steps on your VM and note any differences or extra steps you had to take.

Ans- Step1- Login in Emergency shell

Step 2-Root filesystem read-write mount command: mount -o remount,rw/

Step 3- Edit in Fstab: vi/etc/fstab

Step 4- Delete fake entry

Step 5: Reboot

```
[root@localhost ~]# mount -o rw remount /
mount: /: special device remount does not exist.
mount: (hint) your fstab has been modified, but systemd still uses
the old version; use 'systemctl daemon-reload' to reload.
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# █
```